

FOR DEVELOPERS AND FREE SOFTWARE PEOPLE



A desktop office suite small enough to own.

Built on components from Euro-Office, with a short patch queue and a Python host of under 8,000 lines. For Windows, macOS and Linux.

WHY

The hard part is already free software

UPSTREAM

The document engine

Euro-Office: the editors, the layout engine, the x2t converter, and twenty years of reading other people's .docx. It already runs offline.

OURS

The desktop host

Windows, menus, native dialogs, opening and saving, crash recovery, and packages for three platforms. Small enough for one team to own.

ODF too: .odt, .ods and .odp open and save today, and we are actively improving how faithfully they round-trip.

ARCHITECTURE

The editors talk to one object

The offline build calls a JavaScript object, `window.AscDesktopEditor`, for everything a browser cannot do. Upstream implements it in C++ with CEF and Qt. We implement it in JavaScript and Python.

The editors

sdkjs and web-apps, in the system's own webview: WebKit, WebKitGTK or WebView2

The bridge

`window.AscDesktopEditor`

about 1,000 lines of JavaScript, injected into every page before the editors' own scripts run

The host

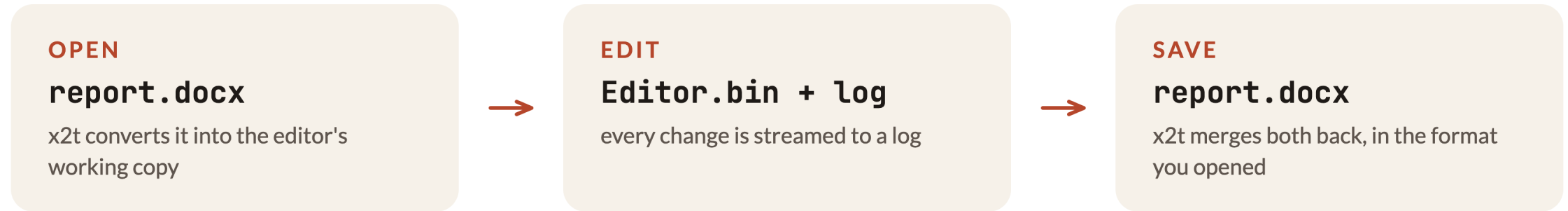
Python, serving the editor on 127.0.0.1: sessions, windows, menus and native dialogs, through pywebview

The converter

upstream's x2t, in C++, for every format; doctrenderer embeds V8 for PDF

HOW A DOCUMENT MOVES

The editors never write a document



Closing saves on its own

The host already holds the working copy and the log, so it saves without asking the editor anything.

A crash loses nothing

The working copy and the log are on disk. Crash recovery offers them back the next time.

Print matches the screen

doctrenderer runs the editor's own layout engine in V8, so the PDF is laid out by the same code.

THE PATCH QUEUE

Pinned and patched in the open

37

patches, across core, sdkjs and web-apps.

None touches the editing engine.

- 21** make it build on macOS
- 8** make it build on Windows
- 7** configuration, branding and build fixes
- 1** fixes an upstream UI bug

Pinned by SHA

Six upstream repositories, each at an exact commit. Bumping a pin is a rebase, done on purpose.

Applied with no fuzz

The queue applies with git am. A patch that stops applying is news, never something to force.

Source for every release

The manifest records each upstream revision and our own commit: AGPL source is a commit plus a patch series.

TAKE PART

Read it, run it, break it

Use it

```
uv tool install libera  
brew install abilian/tap/libera
```

Hack on it

```
git clone https://github.com/abilian/libera-suite  
cd libera-suite && uv sync  
make test
```

330 tests in three tiers, the last driving the real editor in headless Chromium. Host under Apache-2.0, editors under AGPL v3.

How can I help?

- A document that renders wrong, ODF especially
- Flathub, and a one-click .flatpakref
- Windows: code signing, a native ARM64 build
- Keyboard shortcuts in the GTK and Windows menus

Issues: github.com/abilian/libera-suite

Docs: docs.liberasuite.eu/en/develop

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